

Top 20+ Cyber Security Projects for 2026 [With Source Code]

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Cybersecurity has become an integral component of every industry as the world advances technologically. In recent years, an increasing number of young professionals have shown interest in this field. If you are pursuing a course in this field, you should complete a project on cybersecurity as your area of competence.

Beginners with theoretical knowledge should not undertake an impossible endeavor. You should begin with simpler cyber security projects and then gradually progress to the advanced stages. The [best Cybersecurity certification](#) courses will help you get familiar with the latest cybersecurity trends.

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Why Choose Cyber Security?

With the increasing prominence of cybersecurity, you may be curious about what the area comprises and whether it could be the next step in your professional development. Simply reading the news makes it abundantly evident that there is a pressing need for more personnel trained in cyber security and information security projects. Here are some of the most compelling reasons to pursue a career in cybersecurity.

- Professionals in cybersecurity are paid well.
- The variety of specialties has increased.
- Almost every business now prefers cybersecurity.

List of Cyber Security Projects in 2026

Working on independent cyber security projects is one strategy for gaining a competitive advantage. Projects are similar to testimonials that attest to your technical expertise, which distinguishes your profile from the competition. They also facilitate comprehension and implementation of technology. You can also enroll in our [Cybersecurity certification programs](#) to establish more practical skills.

Here are some suggestions for offensive and defensive both kinds of cyber security projects that can assist you in developing your career:

- Test Password Strength
- Integrity Checker
- Simple Malware Scanner Using Yara

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- Bug Bounties and Hackathons
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Cyber Security Projects for College Students

1. Test Password Strength

Source: bitwarden

The password strength project aims to develop a password-strength testing tool. It will provide users with an easy and efficient way to evaluate the strength of their passwords. The tool will analyze various factors such as length, complexity, and inclusion of special characters to determine the strength level. Additionally, it will provide suggestions and tips for creating stronger passwords.

The project will focus on creating a user-friendly interface as a web / Desktop application and incorporating robust algorithms to assess password strength accurately. Ultimately, the goal is to enhance cybersecurity awareness and empower users to protect their accounts with strong passwords.

[Source code](#)

2. Integrity Checker

The Integrity Checker aims to provide security for operating systems. The tool will ensure the integrity and security of system files by verifying their integrity against known hashes or checksums. It will regularly scan the operating system files and compare them with a trusted database to detect any unauthorized modifications or tampering. The integrity checker will provide real-time alerts and notifications for any discrepancies found, enabling administrators or users to take immediate action.

[Source code](#)

3. Simple Malware Scanner Using Yara

Source: joesecurity

The project aims to create a simple malware scanner utilizing the Yara framework. The tool will scan files and directories using Yara rulesets containing known malware patterns and signatures. The scanner will compare the patterns in the rulesets with the

content of the files, allowing for the detection of malicious files and potentially harmful software.

It will provide users with real-time notifications and reports on the presence of any identified malware. The project aims to offer a user-friendly and efficient solution for detecting and mitigating malware threats using the power of Yara's pattern-matching capabilities.

[Source code](#)

4. Bug Bounties and Hackathons

Finding website bugs is another worthwhile endeavor. It can be one of the best cyber security projects for beginners who are interested in making their name in offensive security. There are numerous bug bounty programs on the internet; you can join these programs to obtain practical experience in detecting bugs. Some applications even offer compensation/bounty for finding related bugs.

Participate in hackathons whenever possible. Increasing numbers of companies and platforms are hosting hackathons for prospective cyber security specialists. You can collaborate deeply with graphic designers, project managers, interface designers, and [cyber security domain](#) experts here. Participating in hackathons is a great chance to put your abilities to use and gain a deeper understanding of internet security.

Companies and government agencies are increasing the number of bug bounty programs available, providing more options for security consultants to earn additional money on the side, consider a career shift, or simply take pride in the fact that they found a critical issue in a well-known website.

Once they start reaching milestones, bug bounty hunters and hackathons continue to rise in the ranks. Using their new bug-hunting skills, they can rise in the ranks. An individual's rank is determined by the amount of positive feedback they receive as a result of an increase in the number of hackers who have reported a successful attack in the last 90 days.

[Source Code](#)

Cyber Security Final Year Projects

1. Hashed Password Cracker

Source: arstechnica

The hashed password cracker tool will be designed to crack hashed passwords often used for secure storage and authentication. It will utilize various techniques such as brute force, dictionary attacks, and rainbow table lookups to attempt to reverse-engineer the original password from its hash value.

The cracker will provide a user-friendly interface for inputting hashed passwords and will employ advanced algorithms and optimization techniques to increase efficiency and speed. The project aims to assist users in recovering forgotten passwords or testing the strength of their hashed password implementations.

[Source code](#)

2. Simple Vulnerability Matcher

Source: vuls

As a simple vulnerability matcher tool, it can scan software or systems and compare them against known vulnerability databases to identify potential security weaknesses. It

will leverage a comprehensive database of known vulnerabilities and employ intelligent matching algorithms to identify and prioritize the vulnerabilities found.

The tool will provide users with detailed reports and recommendations on addressing the identified vulnerabilities. The project aims to offer a user-friendly and efficient solution for identifying and addressing security vulnerabilities, enhancing the overall security posture of the scanned software or systems.

[Source code](#)

3. Simple Web Vulnerability Scanner

Source: acunetix

The simple web vulnerability scanner tool will systematically analyze web applications and websites to identify potential security vulnerabilities. It will employ a combination of automated techniques, including crawling, scanning, and fuzzing, to test for common web vulnerabilities such as cross-site scripting (XSS), SQL injection, and insecure direct object references.

The scanner will generate detailed reports outlining the vulnerabilities discovered and recommendations for remediation. The project aims to provide users with an easy-to-use yet effective solution for identifying and mitigating web application vulnerabilities, enhancing the overall security of web-based systems.

[Source code](#)

Cyber Security Projects for Beginners

1. Caesar Cipher Encryption/Decryption

One of the most common and oldest types of encryption techniques is called Caesar Cipher. Julius Caesar mainly employed this technique to communicate with his trusted allies while concealing messages from others.

The Caesar Cipher encryption and decryption can be worked on and learned using a script, making this a challenging cybersecurity assignment. The project is built on the fundamental logic of a numeric key value, which is used to alter the position values of individual alphabets in a text. A key of numeric number “3” is applied to plain text to shift the alphabet to the right three times eventually making the plain text encrypted into the cipher text.

For example if you applied Caesar cipher on the text “SECURITY” it will be converted to cipher text “VHFXULWB”.

The Caesar cipher can be developed using any computer language that can perform encryption and decryption operations on the text provided. This can be the best choice if you want to carry out cyber security projects with [source code](#).

2. Simple Non-hashed Password Cracker

Source: rublon

The simple non-hashed password cracker tool will focus on cracking passwords stored in plaintext or weakly encrypted formats. It will utilize various techniques such as brute force, dictionary attacks, and pattern matching to attempt to reveal the original passwords. The cracker will provide a user-friendly interface for inputting password files or encrypted data and will employ efficient algorithms to optimize the cracking process.

[Source code](#)

3. Simplenetwork Scanner

Source: solarwinds

As a simple network scanner tool, it will scan and analyze networks to discover connected devices, open ports, and services running on those devices. It will provide users with a comprehensive view of their network's topology and identify potential security risks, such as open ports or vulnerable services.

The scanner will offer a user-friendly interface for inputting network ranges or specific IP addresses to scan. It will employ efficient scanning techniques like ICMP, TCP, and UDP to gather information about the networked devices. The project aims to provide users with an easy-to-use yet powerful network reconnaissance and security assessment tool.

[Source code](#)

4. DOS Detection

Source: hindawi

The DoS (Denial of Service) attack detection tool will be specifically designed for beginners. This tool will provide an intuitive interface and simple configuration options to assist users in detecting and mitigating DoS attacks. It will monitor network traffic patterns, analyze abnormal traffic spikes, and identify potential signs of a DoS attack.

The tool will generate real-time alerts and notifications, allowing users to take immediate action to mitigate the attack. The project's objective is to empower beginner-level users with an accessible and effective solution for detecting and responding to DoS attacks, enhancing their systems' overall security and availability.

[Source code](#)

5. SQL Injection

SQL Injection is a sort of Cyberattack in which hackers manipulate the SQL based application queries through input fields available on the front end to fetch data and get login access. This is successful on the web application whose code is vulnerable to SQL injection because of the unavailability of security headers and not securing input fields. In your project you can create a script to check if the application is vulnerable to SQL injection attacks and with respect to the vulnerability script can suggest remediation to close the vulnerability.

[Source Code](#)

Cyber Security Projects for Intermediate

1. Optimized Password Cracker

The optimized password cracker tool will utilize advanced techniques and algorithms to efficiently crack passwords encrypted using various hashing algorithms such as MD5, SHA-1, and bcrypt. It will leverage parallel processing and GPU acceleration to increase the speed and efficiency of the cracking process significantly.

The cracker will also offer customizable options for dictionary, brute-force, and hybrid attacks to cater to different password-cracking scenarios. This project aims to provide users with a high-performance and versatile password-cracking tool that can effectively retrieve lost passwords or evaluate the strength of password implementations.

[Source code](#)

2. Network Anomalies Detection (ML)

Source: mdpi

The network anomaly detection system project will use machine learning techniques to detect anomalies. The system will analyze network traffic patterns and identify abnormal behaviors or activities indicating potential security threats or breaches. It will utilize supervised and unsupervised machine learning algorithms to learn from historical network data and detect deviations from normal network behavior.

The system will generate real-time alerts and notifications, enabling network administrators to mitigate security risks proactively. The project aims to enhance network security by leveraging machine learning to effectively detect and respond to network anomalies.

[Source code](#)

3. File Type Identification Using the Magic Number

Source: hive.blog

This project aims to develop a file-type identification tool using the concept of magic numbers. Magic numbers are unique byte sequences at the beginning of files that can be used to determine their file types. The tool will scan files and analyze their magic numbers to accurately identify their corresponding file formats, regardless of their file extensions.

It will maintain a comprehensive database of magic numbers and their associated file types. The tool will provide users with quick and reliable identification of unknown file types, aiding in file management, data recovery, and cybersecurity efforts. Advanced features should include identifying files with mixed identification or embedded files.

[Source code](#)

4. Keylogging (Spyware)

In computing, a keylogger is a piece of software that monitors all the keystrokes that take place on a computer's keyboard. It monitors previously pressed and currently active keys to record user activity. You can consider the following project on the concept of keylogging:

- Create a script that can record keystrokes.
- Create and bind a keylogger with legitimate application/software to record user activity, basically spyware.
- Create a script that can detect the presence of a keylogger on endpoint.

[Source Code](#)

This project is ideal for those who are interested in offensive security. If you find yourself interested in this, do check such projects offered in [CEH certification training](#).

5. Cloud Access Security Broker (CASB)

For businesses that have previously deployed several SaaS apps, CASBs give a visibility and administrative control point. Using a cloud application discovery to uncover hidden IT resources can help validate this type of project.

It is possible for leaders to assess whether their organization has visibility and control over sensitive data utilized and shared by SaaS apps and determine the level of visibility and control required for each cloud service. Contracts focusing on the discovery and security of sensitive data should be short-term.

[Source Code](#)

6. Lost Data Retrieval

Malware can corrupt, destroy, or distort data, making data recovery abilities crucial to [cyber incident response](#). Ransomware attacks encrypt a victim's data and demand money in exchange for decryption. This can be a good addition to your career as it involves information security in project management.

A ransomware data recovery technique can be used to train data retrieval skills. Concentrate on recovering impacted systems from backups. Next, develop a strategy for extracting corrupted or destroyed data from storage devices using data recovery tools.

[Source Code](#)

Advanced Cyber Security Projects

1. Advanced Network Scanner Using Nmap

Source: [online-tech-tips](#)

The advanced network scanner will utilize the Nmap scripting engine. The project will leverage the powerful capabilities of Nmap and its extensive library of scripts to perform comprehensive network scans and assessments. It will support various scanning techniques, including host discovery, port scanning, service enumeration, and vulnerability detection.

The scanner will utilize custom or pre-existing Nmap scripts to automate complex tasks and gather detailed information about networked devices and services. This project aims to provide users with an advanced and customizable network scanning solution, enabling in-depth network analysis and enhancing overall security posture.

[Source code](#)

2. Advanced Network Packet Capturing Tool

The tool will enable the capture and analysis of network packets to gain insights into network traffic and diagnose network-related issues. It will provide features such as real-time packet capturing, filtering, and detailed packet analysis.

The tool will support various protocols and offer advanced filtering options to focus on specific network traffic of interest. Additionally, it will provide visualization capabilities and export options for further analysis. The project aims to provide network administrators and analysts with a robust and feature-rich tool to monitor, troubleshoot, and optimize network performance.

Instead of reinventing the wheel and writing everything from scratch, libraries and frameworks in different programming languages can be used to develop a customized packet-capturing tool.

[Source code](#)

3. Exploit Development: Metasploit

Source: [varonis](#)

The project focuses on providing methods and features for automating the exploit development using the functions and tools available in the popular Metasploit framework. Metasploit is a powerful and widely used penetration testing tool that identifies and exploits vulnerabilities in target systems.

The project will involve learning and utilizing the Metasploit framework to develop custom exploits, payloads, and modules. It will explore techniques such as remote code

execution, privilege escalation, and post-exploitation activities.

The goal is to enhance understanding and proficiency in exploit development, enabling security professionals to effectively assess and strengthen the security of target systems, networks, and applications.

[Source code](#)

4. Packet Sniffing

Security specialists in the industry frequently employ this technique to keep tabs on how data is transmitted across their network. Typically, a packet comprises the information or data that is to be transported between two network sites, from the sender to the intended recipient.

This technique allows us to track and monitor the transmission of data packets from the source to the destination. You can do an outstanding final year thesis if you will consider this project, Network security projects primarily deal with network's intrusion detection, monitoring illegal access and modification.

When working with this cybersecurity project, you can either use Python scripts to follow the information in each packet or focus on its source and destination. Additionally, you can set up a system to trace any unauthorized access to critical information or ensure that the network infrastructure surrounding this packet transmission is protected from external incursions.

[Source Code](#)

5. Wazuh Open Source SIEM

Wazuh is nowadays popularly used by businesses as SIEM solution in their Security operation center. It is completely packed with threat detection, integrity monitoring, and incident management features. Wazuh collects, aggregates, indexes, and evaluates security data, enabling enterprises to identify malicious risks and potential behavioral anomalies. Among its many features are the following:

- Intrusion Discovery
- Log Data Analysis
- File Integrity Observation
- Vulnerability Discovery

Wazuh is an open-source solution, so its full access is available free of cost without any requirement of licenses, along with an amazing supportive community. Those who are interested in defensive security should implement Wazuh in their lab environment and experience its usage, which will surely distinguish their resume from that of others in defensive security.

[Source code](#)

6. Cloud Security Posture Management

This can also be one of the great and unique cybersecurity thesis topics if you want to stand out in the competition. Extremely dynamic cloud applications necessitate an automated DevSecOps approach to security. Organizations must establish IaaS and PaaS-wide standard controls and allow automatic evaluation and repair.

[Source Code](#)

Conclusion

Information leaks, data robberies, and a wide variety of other dangers can all be avoided with the use of an excellent piece of technology known as cyber security, which secures

computer systems and networks. Cyber security projects are necessary for getting practical experience and improving a candidate's credibility in preparation for a potential job.

You can improve your chances of landing a big job by participating in many cybersecurity-related projects. [KnowledgeHut ITIL training online](#) course will help you gear up for a highly-rewarding IT career.

Frequently Asked Questions (FAQs)

1. What are the types of cyber security? —

There are a few general categories of cyber security, which can be applied to a wide range of scenarios, from enterprise to personal computing.

- Network security
- Information Security
- Application security
- Cloud Security
- EndPoint Security

2. Is cyber security a good career? —

There's never been a better time to start a career in cybersecurity, thanks to the increased need for qualified workers. According to the Bureau of Labor Statistics, information security analysts are expected to have a 31% increase in employment between 2019 and 2029.

3. What are IT security projects? —

IT security aims to prevent unauthorized third parties from accessing and manipulating data and systems. Protecting information and particularly analyzing information is the goal of IT security. IT security projects are those that deal with this specific area of security. Examples of some projects are:

- Antivirus
- SIEM Solution
- Firewalls
- Vulnerability Scanners

4. What do you mean by Keylogger in cyber security? —

Hackers can access your personal information by using keyloggers and software applications monitoring your online activities. Keyboard strokes can be used to track everything you do online, including the passwords and credit card details you enter. The software records every keystroke you make.

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